

# Mid-Progress Report

GitHub Repository: [XAI-Project](#)

## 1.1 Data Preprocessing

The following preprocessing steps were applied:

- Loaded dataset using pandas
- Dropped unnecessary column (id)
- Encoded categorical variables using **Label Encoding**
- Applied **StandardScaler** for feature scaling

### Handling Imbalanced Data:

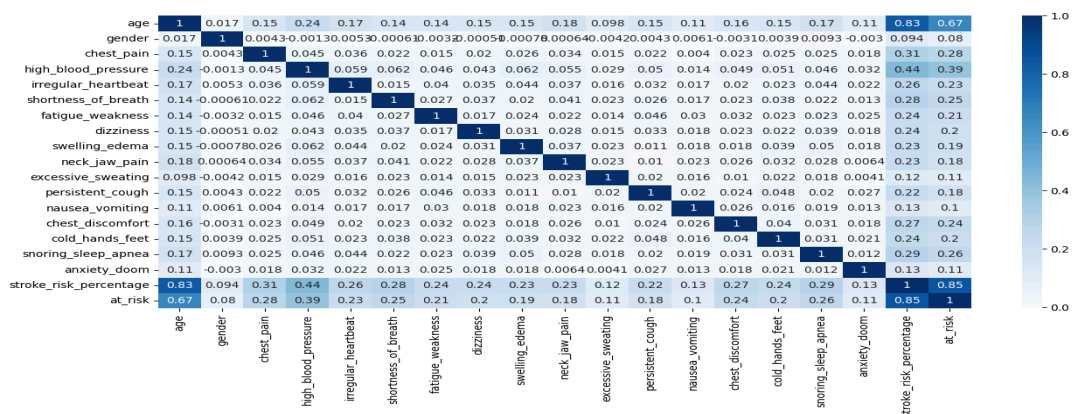
The dataset was highly imbalanced, so **Upsampling** was applied:

- Minority class (at\_risk = 1) was resampled

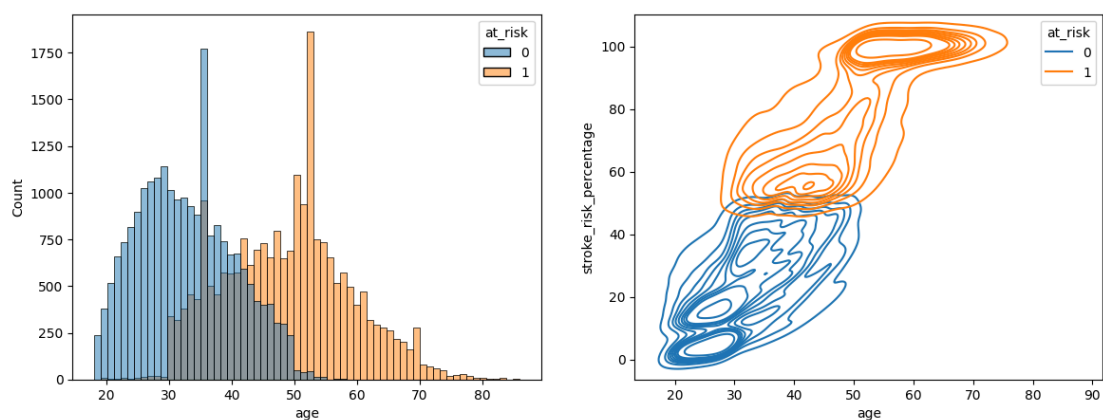
## 1.2 EDA

Several metrics and visualizations were implemented to analyze the data. The heatmap showed that age is highly correlated with the risk and percentage of stroke while chest pain and high blood pressure were moderately correlated. While examining the effect of age on stroke, beyond the obvious positive correlation (age increases, risk increases), it was found that there is a region where age alone can't separate between the at risk and the not at risk well, which suggests that in that region some other features must be used. Some outliers were also found in the age category relating to the risk (high age – not at risk / low age – at risk), but they seemed to be realistic cases so they were left in the data. Some distribution insights were derived from the bar plots such as the frequency of the presence of a certain feature (anxiety doom is quite rare but when present it is more likely to be at risk for stroke). Regarding chest pain and high blood pressure, they were both present more in cases with higher stroke risk percentage. It was also found that high blood pressure is more strongly associated with stroke risk, such that high blood pressure but no chest pain has a higher stroke risk than having chest pain but no high blood pressure.

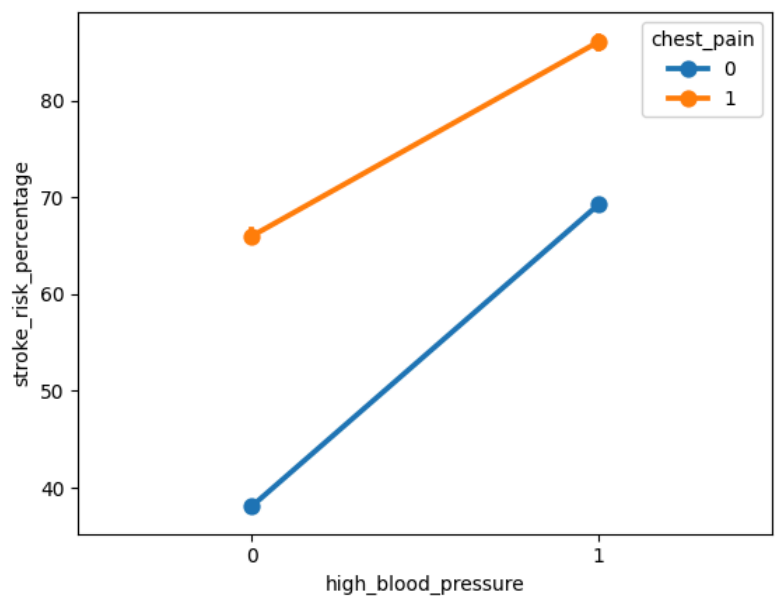
Heatmap:



Effect of age on stroke risk:



Interaction of chest pain and high blood pressure with stroke risk:



### 1.3 Feature Engineering / Feature Selection

Feature engineering and feature selection techniques were performed at this stage of the project in order to identify the most relevant predictors of stroke risk. Its objective was to reduce the dimension, improve the model performance and keep only the most informative features.

#### 1.3.1 Feature Engineering:

A new feature '**symptom\_count**' was created by summing all symptom related columns for each patient. This feature may be useful in representing the overall symptom burden in a single value, which may improve the ability of the model to capture the relationship between symptoms and stroke risk.

#### 1.3.2 Feature Selection

- **Correlation Analysis**

A correlation analysis was performed between all features and the target variable (stroke\_risk\_percentage). This step helped identify linear relationships and provided an initial understanding of which variables are most associated with stroke risk.

- **Select K-Best**

Select K-Best was applied to select the top 12 features based on their statistical relationship with the target variable. This method evaluates how strongly each feature separates the classes in the dataset.

- **Recursive Feature Elimination**

RFE with Logistic Regression was used to iteratively remove less important features and keep the most significant 12 features. This method relies on model performance to rank feature importance.

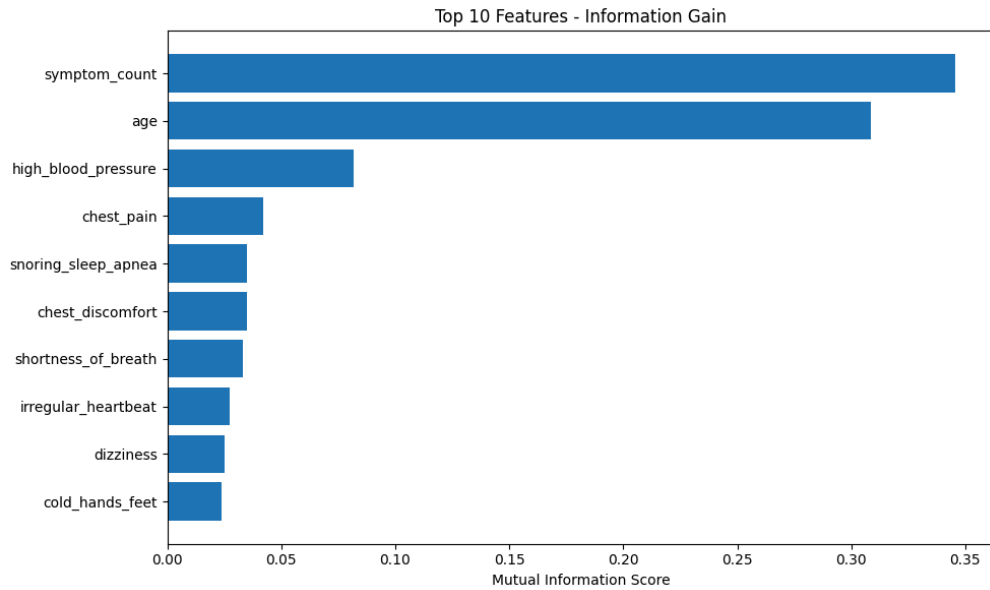
- **Information Gain**

Mutual Information was used to capture both linear and non-linear relationships between features and the target. The top 10 features were selected and ranked based on their information gain scores.

#### Final Feature Selection

The final set of features selected was based on common features appearing in most of the feature selection methods used in the study (SelectKBest, RFE, and Mutual Information). This way only the most stable and consistently important features remained. Selected Features: ['age', 'chest\_pain', 'high\_blood\_pressure', 'irregular\_heartbeat', 'shortness\_of\_breath', 'fatigue\_weakness', 'dizziness', 'neck\_jaw\_pain', 'chest\_discomfort', 'snoring\_sleep\_apnea']

These features were chosen as the most related to predict the risk of stroke and were used in the final dataset to train the model.



## 2. Model Implementation

### 2.1 Model Used: XGBoost Classifier

#### Why XGBoost?

- High performance on tabular data
- Handles non-linear relationships
- Works well with imbalanced datasets

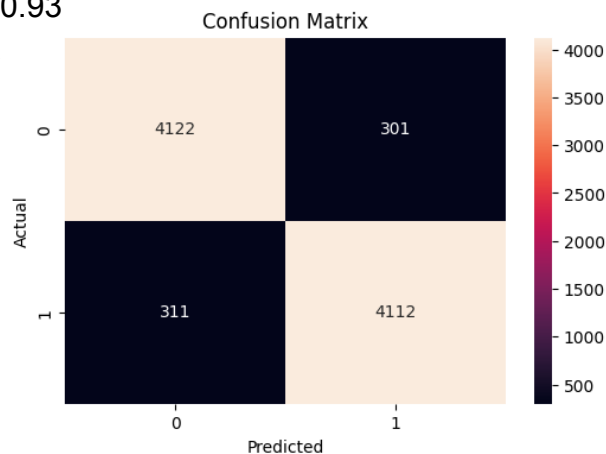
#### Training Process:

- Data split into training and testing sets
- Model trained on selected features
- Predictions made on test set

## Model Evaluation

### Metrics Used:

- Accuracy Score (0.9308161881076192 )
- Classification Report
  - Precision: 0.93
  - Recall: 0.93
  - F1-score: 0.93
- Confusion Matrix



## Observations:

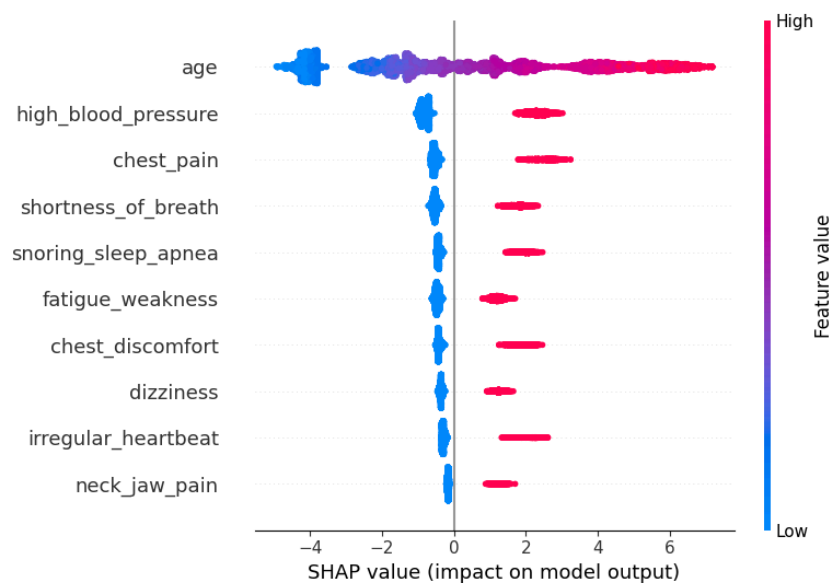
- The model successfully distinguishes between classes after balancing
- Confusion matrix shows improved detection of minority class
- Performance improved compared to unbalanced data

## Model Interpretability

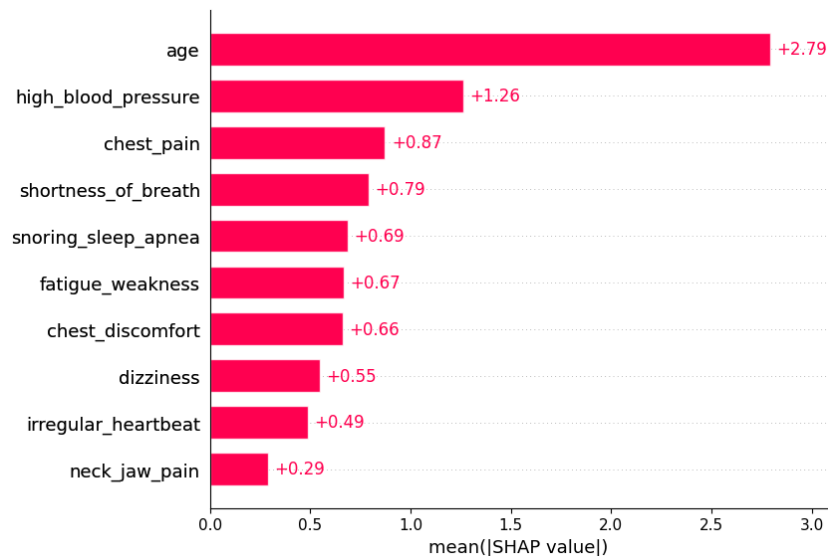
Interpretability was applied using **SHAP (SHapley Additive Explanations)**:

### Techniques Used:

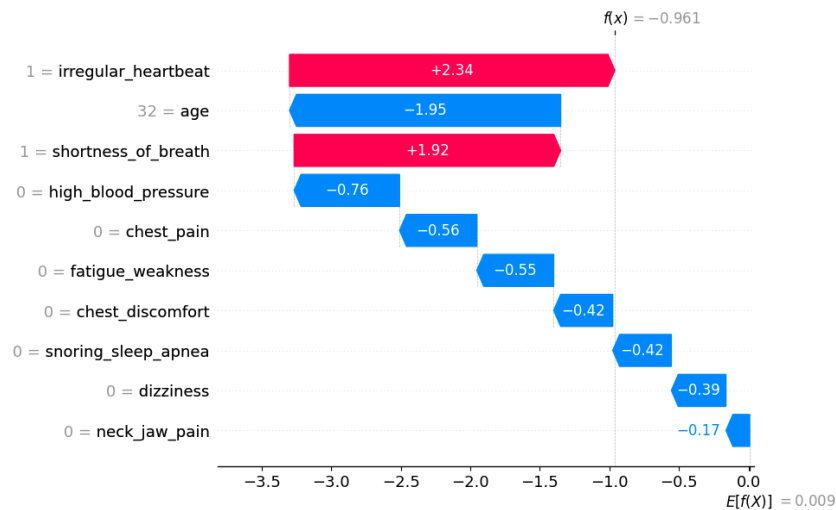
- SHAP Summary Plot



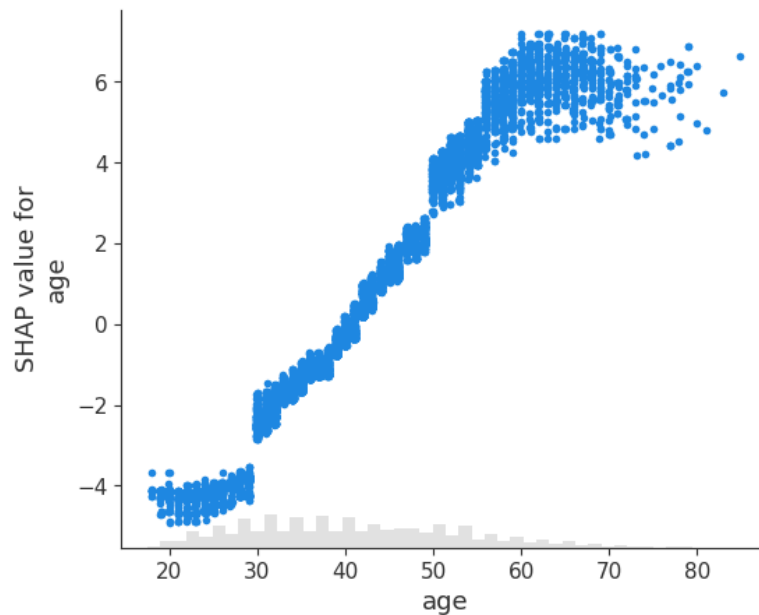
- SHAP Feature Importance (Bar Plot)



- SHAP Waterfall Plots (Local explanations)



- SHAP Dependence Plot



### Key Insights:

- **Age** proved to be the most correlated feature with stroke risk, signifying the fact that patients of an advanced age are much more susceptible to being identified as at risk.
- **Chest pain, high blood pressure, and irregular heartbeat** were found to be critical cardiovascular predictors that helped forecast the probability of stroke.

- Such factors as shortness of breath, fatigue, weakness, dizziness, and pain in the neck or jaw also correlated well with the target variable.
- A number of attributes were only moderately connected to the stroke risk factor and not highly correlated; this means that multiple symptoms should be considered in the prediction process.
- There were many instances of the minority class in the initial dataset, leading to a skewed proportion of classes.
- After equalizing classes within the dataset, it became easier to predict the target class.

## Challenges Faced

In this stage of the process, a number of difficulties arose during data preparation, modeling, and interpretation.

One of the key issues that appeared in the course of work was the class imbalance in the initial dataset. There were significantly fewer patients classified as being at risk of stroke than not-at-risk patients. Class imbalance leads to the biasing of the learning algorithm toward the more numerous classes. Upsampling was performed to address this issue and balance the data.

Another problem was the difficulty in interpreting the results of XGBoost predictions, as the model does not provide any information on how it arrived at its predictions. Therefore, SHAP values were computed to explain the contribution of each feature in the prediction both globally and locally.

## 2.2 Random Forest (RF)

The RF model (regression – stroke risk percentage) was trained on the training data (80%) with the same parameters that were used in the original paper [1]. It was then tested on the test data (20%) with the following results:

**Mean Squared Error (MSE):** 48.1759621842011

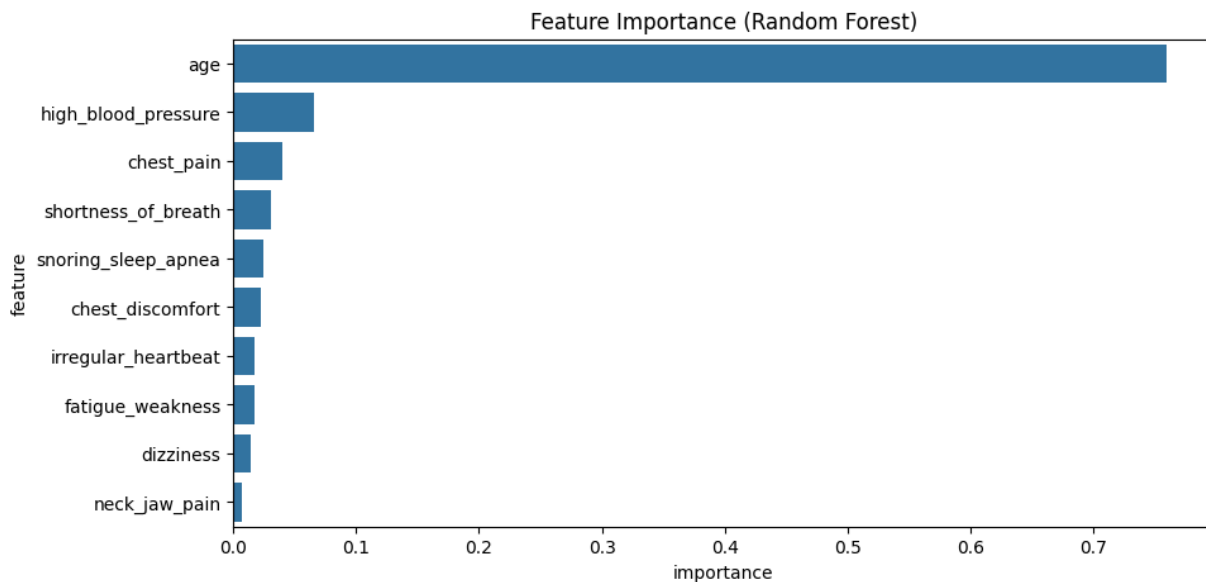
**Mean Absolute Error (MAE):** 5.024658517435232

**R-squared (R2):** 0.9497152032369891

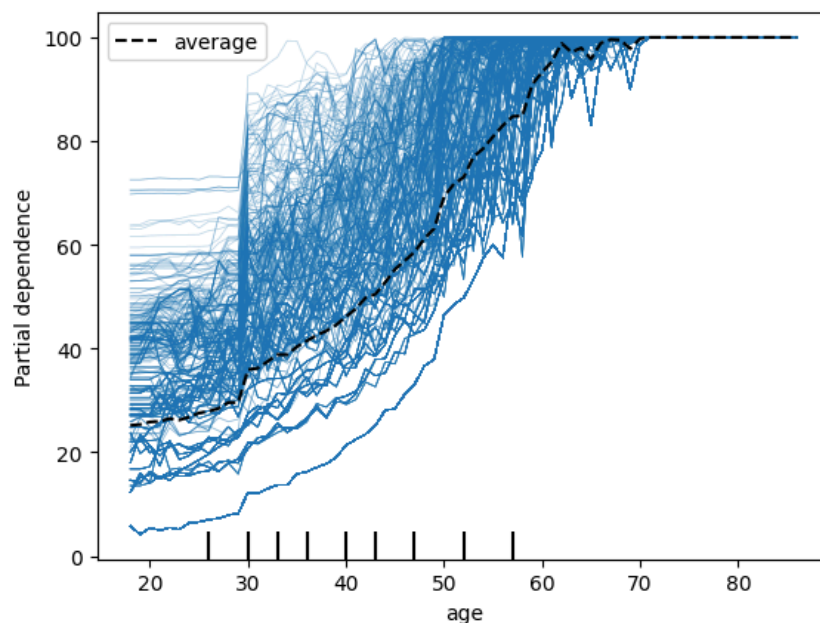
Several interpretability techniques were used to explain how the model worked and its results.

First, the CART algorithm (variance reduction) showed that age had the highest influence on the results by a huge margin and after that were high blood pressure and chest pain in that order (as observed in the EDA). After trying to visualize some trees from the RF, it was found that the trees were too complex (and the data too big) to gain any insights from.

CART algorithm:



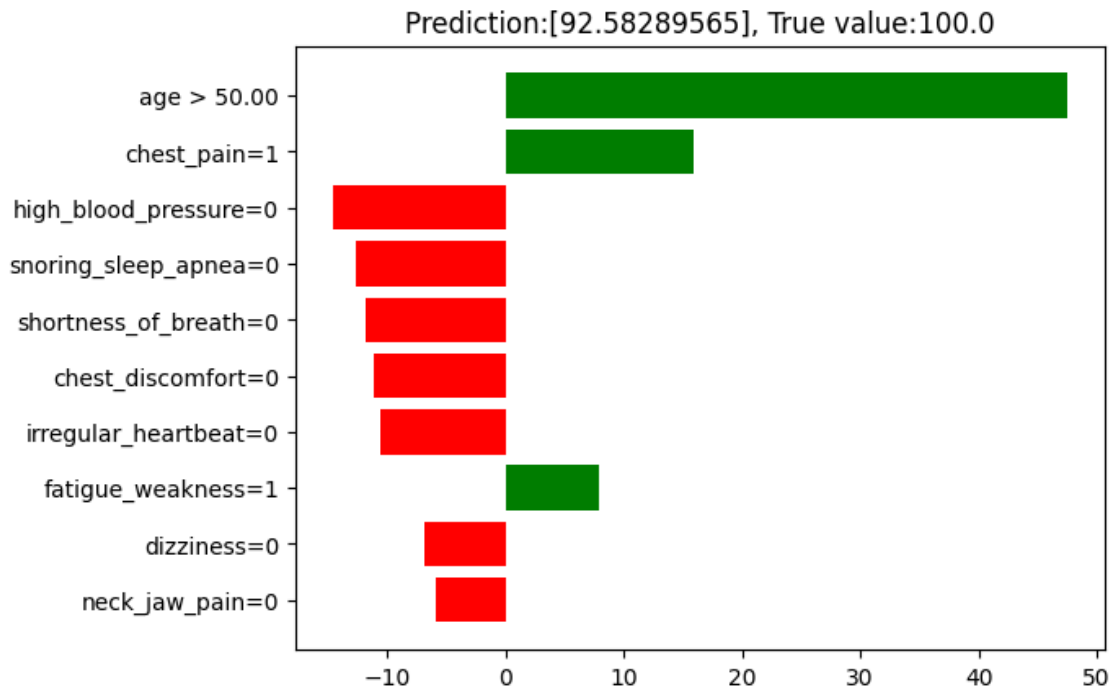
Permutation importance and LOFO produced the same results as the CART algorithm. The partial dependence plot (PDP) and the ICE plot showed that age had a direct impact on the risk percentage (a positive correlation).





LIME and SHAP were also used to explain specific instances. A summary plot from SHAP (500 samples) also supported the results from the other techniques (age had the most impact, etc...).

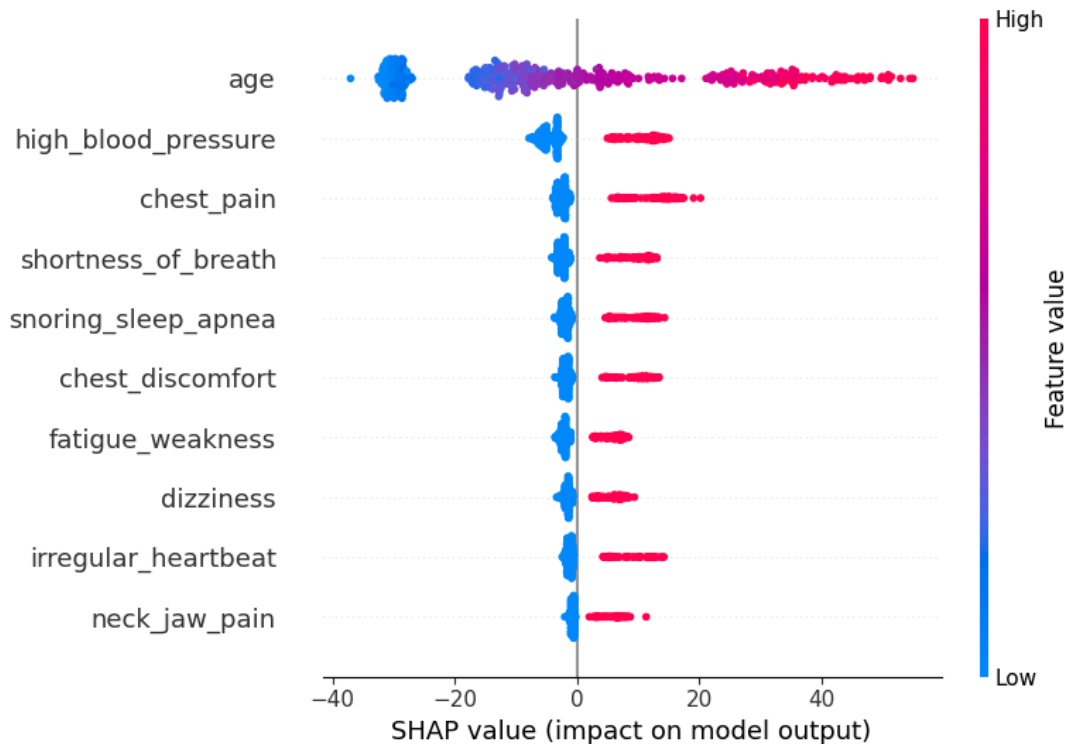
LIME:



SHAP:



SHAP Summary Plot (500 samples):



To summarize, the results were consistent across all the interpretability techniques, they showed that age influenced the model's decision the most with high blood pressure and chest pain coming next.

## 2.3 Staking Model

The Stacking Regressor was built to predict stroke risk percentage by combining multiple regression models. The dataset was split into training (75%) and testing (25%) sets to ensure proper evaluation. Four baseline models were implemented: Linear Regression, Decision Tree, Random Forest, and K-Nearest Neighbors (KNN).

The models were evaluated using three main metrics: MAE (Mean Absolute Error), MSE (Mean Squared Error), and  $R^2$  score.

- Linear Regression showed stable but relatively simple performance, as it assumes a linear relationship between features and target showing these results:
  - MAE: 7.733256816552589
  - MSE: 101.06108355936075
  - R2: 0.8955640985128858

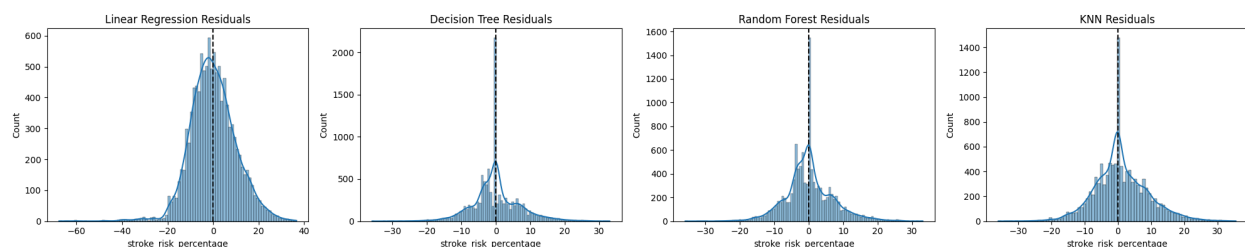
- Decision Tree captured nonlinear patterns but tended to overfit, leading to less generalization.
  - MAE: 4.964332749842316
  - MSE: 49.47070958083346
  - R2: 0.9488772733250295
- Random Forest performed better overall due to ensemble learning, reducing variance and improving prediction accuracy.
  - MAE: 5.039173862974332
  - MSE: 48.66787734379162
  - R2: 0.9497069152155118
- KNN performance depended on distance-based similarity, but it was sensitive to noise and feature distribution.
  - MAE: 5.981320430496519
  - MSE: 65.90318013927828
  - R2: 0.9318960594295379

Among these, Random Forest achieved the best balance between bias and variance, making it the strongest individual model.

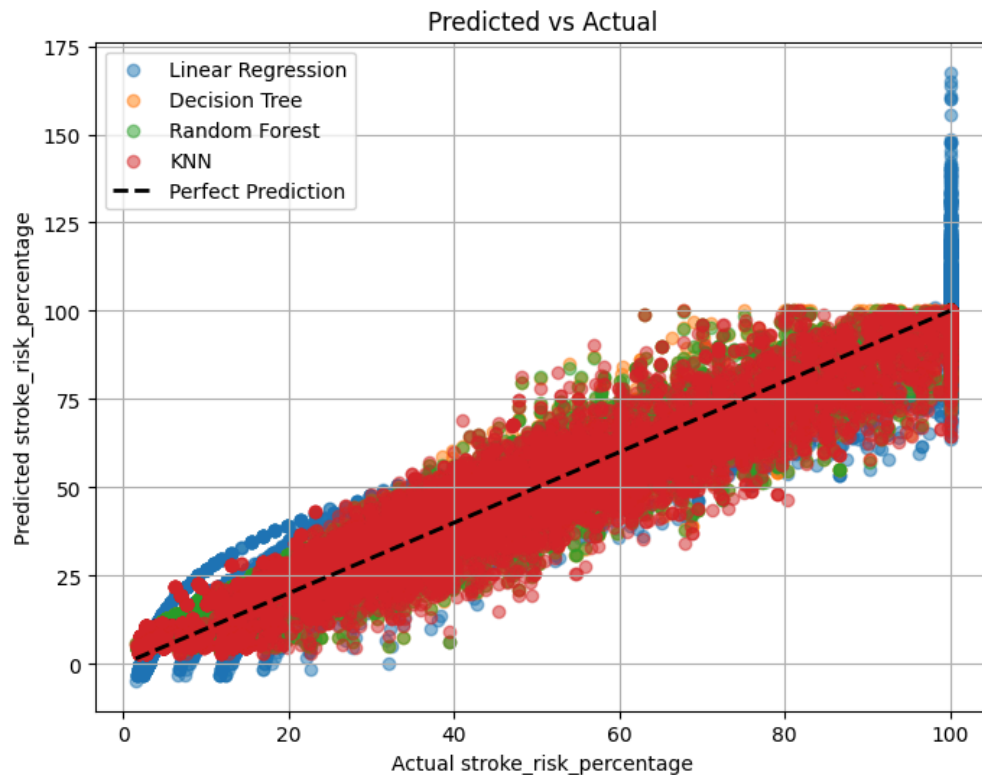
The stacking model achieved improved performance compared to most individual models, demonstrating the effectiveness of model combination with the following results:

- MAE: 5.162300719334152
- MSE: 48.41134294690611
- R2: 0.9499720162816947

## Residual plots for each model:



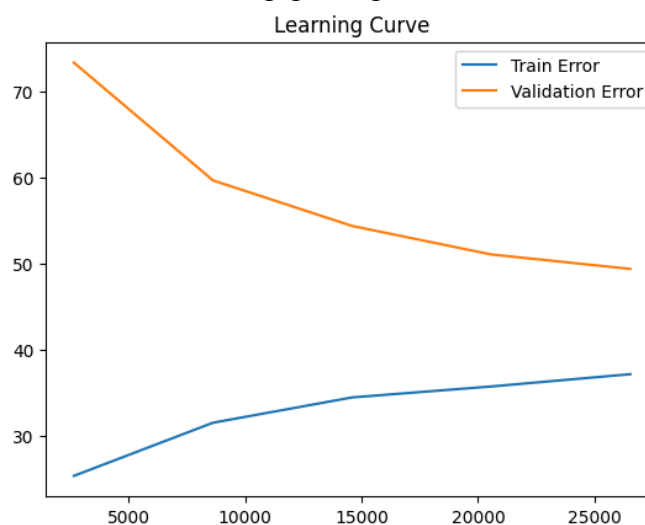
## Predicted VS Actual:



GridSearchCV was used to find the optimal set of hyperparameters in order to maximize the performance of the Random Forest, Decision Tree, and KNN models. This procedure concentrated on important factors that directly affect the accuracy and generalization of the model, such as tree depth, the number of estimators, and the number of neighbors. The models were adjusted to improve predicted accuracy by utilizing cross-validation to assess various configurations. The Random Forest model maintained to perform better than the others after tuning, confirming its robustness.

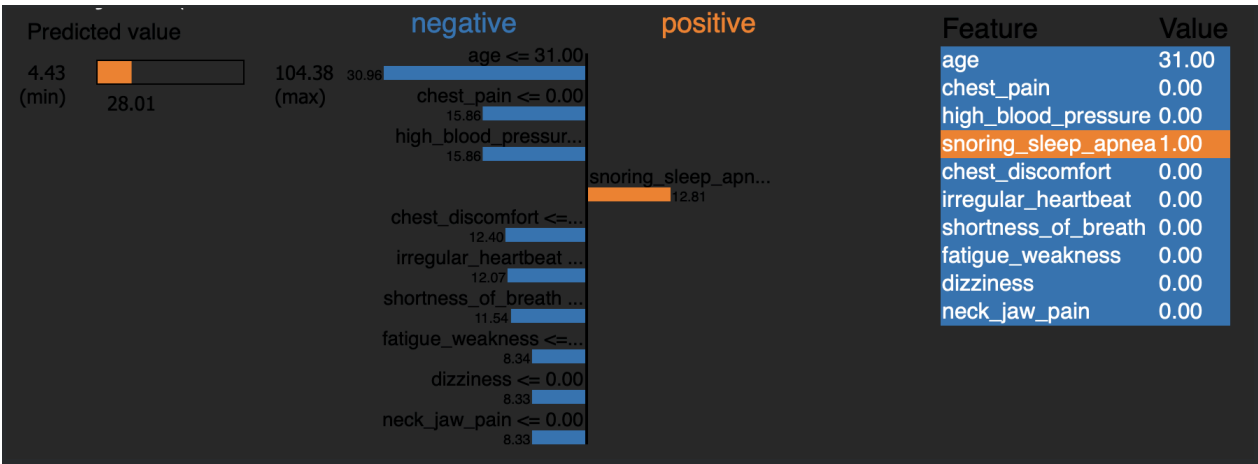
## Learning Curve

The learning curve compares training error with validation error to assess model performance. The Random Forest model shows a reasonable gap between the training error and high validation error, indicating good generalization with slight room for improvement.



Explainable AI Techniques

LIME was used to explain predictions for individual instances by providing local interpretations of the model’s decisions. It highlights which features contribute to increasing or decreasing the predicted stroke risk for a specific patient.



SHAP was applied to show the direction of each feature’s impact on the model output.



## References:

- [1] K. Moulaei *et al.*, "Explainable artificial intelligence for stroke prediction through comparison of deep learning and Machine Learning Models," *Scientific Reports*, vol. 14, no. 1, Dec. 2024. doi:10.1038/s41598-024-82931-5.
- [2] Sorayaie Azar, A., Rikan, S. B., Naemi, A., Bagherzadeh Mohasefi, J., Pirnejad, H., Bagherzadeh Mohasefi, M., & Wiil, U. K. (2024). Application of machine learning techniques for predicting survival in stroke patients. *European Journal of Medical Research*, 29, 547. <https://doi.org/10.1186/s40001-024-02147-1> .
- [3] S, S., Chadaga, K., Sampathila, N., Prabhu, S., Chadaga, R., & S, S. K. (2023). Multiple explainable approaches to predict the risk of stroke using artificial intelligence. *Information*, 14(8), 435. <https://doi.org/10.3390/info14080435> .